Request for Proposal (RFP) - Data Analytics System

Data Lake House Implementation on AWS for Advanced Analytics

Project developed for

Mexican Football Federation

Implemented by

Team 6 - Data Engineering, UNAM 2026.

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1. Introduction and Background

The coaching staff of the Mexican National Team is interested in modernizing their sports data processing and analysis capabilities. The goal is to establish a robust infrastructure capable of transforming raw international match data into strategic information to support tactical decision-making and the analysis of potential rivals ahead of the FIFA 2026 World Cup.

With the increasing availability of statistical information in international football, there is a need to implement a data engineering platform capable of automating the collection, storage, processing, and querying of historical data from national teams. The primary goal is to generate reliable, centralized metrics that facilitate sports scouting and comparative performance analysis between teams.

To achieve this, the development of an architecture based on AWS cloud services is proposed, using tools such as Amazon EC2, Amazon S3, AWS Glue, and Amazon Athena to build a scalable Data Lake oriented toward football analytics.

2. RFP Objective

The purpose of this Request for Proposal (RFP) is to identify and select a technology partner or consulting team for the design, development, and implementation of a robust cloud-based Data Lake, oriented toward the analysis of sports data from national football teams.

The solution must enable the automation of ingestion, storage, transformation, and querying processes for historical international match data, using scalable and cost-effective cloud services. It also seeks to build a platform capable of generating strategic metrics and sports statistics to support tactical analysis, scouting, and evaluation of potential rivals of the Mexican National Team ahead of the FIFA 2026 World Cup.

The project involves the use of AWS technologies to ensure availability, scalability, and operational efficiency, as well as the implementation of ETL processes and analytical queries that facilitate centralized exploitation of sports data.

3. Required Technical Scope

The proposal must address the following technical pillars in detail:

Data Architecture

Design of a Data Lake House architecture capable of supporting structured sports datasets obtained from external APIs, enabling efficient storage, organization, and processing of historical international football match data. The solution should include a layered architecture (Bronze, Silver, and Gold) to ensure data traceability and quality.

Security and Governance

Implementation of access control policies using AWS IAM, service-level permission management, and data protection through encryption both at rest and in transit. The proposal should also include proper dataset organization and governance mechanisms to facilitate administration and analytical access.

Scalability

The proposed infrastructure must be capable of scaling elastically according to increases in data volume, number of analyzed national teams, or complexity of analytical queries required by the Mexican National Team coaching staff.

Automation

Automation of extraction, transformation, and loading (ETL) processes, as well as analytical query generation and historical dataset updates. The solution should allow consistent and maintainable deployments through reusable scripts and automated workflows.

Data Processing and Analytics

Implementation of data cleaning, validation, and transformation processes using distributed technologies such as AWS Glue and PySpark. Additionally, the platform should enable statistical analysis through SQL queries in Amazon Athena to generate strategic football performance metrics.

Cloud Integration

Use of AWS cloud services such as Amazon EC2, Amazon S3, AWS Glue, and Amazon Athena to build a centralized, flexible, and cost-efficient solution.

4. Proposal Requirements

Participants must include the following sections in their proposal:

Section	Expected Content
Team Profile	Previous experience in data engineering projects, cloud architectures, and sports analytics solutions. Experience with AWS services, ETL automation, and analytical processing will be highly valued.
Work Methodology	Description of the methodology used for project development, including analysis, ingestion, transformation, validation, and data exploitation phases. Agile approaches such as Scrum or Kanban may be used.
Proposed Architecture	Presentation of the proposed technological architecture, including AWS services such as Amazon EC2, Amazon S3, AWS Glue, and Amazon Athena, as well as the data flow between the Bronze, Silver, and Gold layers of the Data Lake.
Data Processing	Explanation of the ETL processes used for cleaning, normalization, validation, and transformation of historical international football match data.
Security and Governance	Strategy for access control, IAM permission management, and protection of cloud-stored datasets.
Budget and Timeline	Estimated breakdown of infrastructure, storage, and processing costs, along with a detailed implementation and testing schedule.

5. Evaluation Process

Proposals will be evaluated according to the following criteria:

Evaluation Criteria	Description
Technical understanding of the proposed challenges (40%)	Level of understanding regarding the project requirements, including Data Lake architecture, sports data processing, ETL automation, and statistical analysis focused on football scouting and opponent analysis.
Previous experience and proven success cases (30%)	Previous experience in data engineering projects, AWS cloud solutions, analytical processing, and development of sports analytics platforms or similar systems.
Total implementation cost (20%)	Evaluation of infrastructure, storage, processing, and operational maintenance costs associated with the proposed solution.
Estimated delivery time (10%)	Feasibility and clarity of the implementation schedule, including development, testing, validation, and deployment phases.

6. Submission Instructions

The proposal must be submitted in PDF format and addressed to the coaching staff of the Mexican National Team before the deadline established in the academic project schedule.

The document must include all technical, operational, and financial information related to the proposed solution, as well as architectural diagrams, implementation timelines, and descriptions of the technologies used.

Additionally, the proposal is expected to include scalability, automation, and sports analytics strategies that strengthen scouting capabilities and opponent evaluation processes for the FIFA World Cup 2026.